

Project Executable Files

Date	1 July 2026
Team ID	SWTID-2026-9097
Project Name	Voice Based Concept Understanding Analyzer
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes

10	Demo video or walkthrough (if required)	Yes
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Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```
VBCUA/
|
|  └─ backend/
|    |  └─ app.py          # FastAPI backend application
|    |  └─ routes.py       # API routes
|    |  └─ services/       # Business logic
|    |  └─ utils/          # Helper functions
|    └─ requirements.txt   # Backend dependencies
|
|  └─ frontend/
|    |  └─ app.py          # Streamlit frontend
|    |  └─ pages/          # Streamlit pages
|    |  └─ utils/
|    |    |  └─ speech_to_text.py
|    |    |  └─ transcript.py
|    |    |  └─ concept_extractor.py
|    |    └─ summarizer.py
|    └─ assets/
|
|  └─ tests/                # Test cases
|
|  └─ reference_concepts/   # Reference concept files
|
|  └─ uploads/              # Uploaded audio files
|
|  └─ README.md             # Project documentation
|  └─ requirements.txt      # Python dependencies
└─ .gitignore
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	NA

Login Credentials (Demo)	NA
Platform / Hosting Provider	[e.g., Render, Vercel, AWS, Heroku, Azure]
Repository Link	https://github.com/Kirti3005/Voice--Based--Concept--Understanding--Analyzer--SWTID-2026-9097
Demo Video Link	NA

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Prerequisites

- Python 3.10 or later
- Git
- pip package manager

Steps

1. Clone the repository

```
git clone https://github.com/Kirti3005/Voice--Based--Concept--Understanding--Analyzer--SWTID-2026-9097.git
```

2. Navigate to the project folder

```
cd Voice--Based--Concept--Understanding--Analyzer--SWTID-2026-9097
```

3. Create a virtual environment

```
python -m venv venv
```

4. Activate the virtual environment

```
venv\Scripts\activate
```

5. Install required dependencies

```
pip install -r requirements.txt
```

6. Start the FastAPI backend

```
uvicorn backend.app:app --reload
```

7. Open a new terminal and start the Streamlit frontend

```
streamlit run frontend/app.py
```

8. **Open the generated localhost URL (typically)**

<http://localhost:8501>

9. **Upload an audio file** :- To analyze concepts and view the generated transcript, summary, and concept analysis.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Accuracy depends on the quality of the uploaded audio.	Use clear, noise-free recordings for better transcription accuracy.
2	Large audio files may take longer to process.	Compress or split lengthy audio files before uploading.
3	Application currently runs only on localhost and is not publicly deployed.	Can be deployed later using platforms such as Render, Railway, or Azure.